RW–11037

Sub. Code
545553/ 541553

M.C.A. (R)/M.C.A. (WE) DEGREE EXAMINATION, APRIL 2016

SOFTWARE TESTING METHODOLOGIES

(Common of M.C.A. (R)/M.C.A.(WE))

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** the questions.

1. What is the purpose of software testing?

2. State the applications of path testing.

3. Differentiate between domain and interface testing.

4. What are the techniques used in transaction flow testing?

5. List out the applications of regular expressions.

6. What is decision table?

7. Define bugs.

8. What are state graphs?

9. Give any two tools for testing.

10. What is meant by synchronization of test cases?

Section B (5 × 5 = 25)

Answer **all** the questions, choosing either (a) or (b).

All questions carry equal marks.

11. (a) Write a note on purpose of testing.

Or

(b) Give a brief note on flow graphs in testing.

12. (a) Discuss about path testing with suitable example.

Or

(b) Explain the domain testing.

13. (a) Write a short note on path products and path sums.

Or

(b) Explain logic based testing.

14. (a) Compare and contrast good state and bad state graphs.

Or

(b) Write a short note on issues in state testing.

15. (a) Explain rapid testing.

Or

(b) Discuss about performance testing on any real time database application.

Section C

(3 × 10 = 30)

Answer any **three** questions.

All questions carry equal marks.

16. Explain the concept of path testing and its applications.

17. Discuss in detail about data and flow testing.

18. Explain about regular expressions and flow anomaly detection.

19. Discuss in detail about graph matrices and its applications.

20. Write a note on functional and regression testing using winrunner tools.

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RW–10778

Sub. Code
541202/ 545202

M.C.A.(R)/(WE) DEGREE EXAMINATION, APRIL 2016

Second Semester

OPERATING SYSTEMS

(Common for M.C.A.(R)/WEP)

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. Define volatile and nonvolatile storage.
2. Define distributed system.
3. List out the process state.
4. Define preemptive and non-preemptive scheduling.
5. Define semaphore.
6. Write about deadlock prevention.
7. Define swapping.
8. Define first fit.
9. What is the use of raw disk?
10. What is constant angular velocity?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

All questions carry equal marks.

11. (a) Explain about I/O structure.

Or

(b) Describe the operating system services.

12. (a) Describe the operations on process.

Or

(b) Explain about scheduling criteria.

13. (a) Describe the usage of semaphores.

Or

(b) Explain about recovery from deadlock.

14. (a) Discuss about basic concept of paging.

Or

(b) Describe the file system mounting.

15. (a) Describe the overview of directory.

Or

(b) Explain about disk management.

Part C (3 × 10 = 30)

Answer any **three** questions.

All questions carry equal marks.

16. Explain the structure of OS.

17. Describe the shortest job first scheduling with an example.

18. Describe the classic problems of synchronization.

19. Explain about contiguous memory allocation.

20. Describe the different types of allocation methods.

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RW–10779

Sub. Code
541203/ 545203

M.C.A. (R)/WE DEGREE EXAMINATION, APRIL 2016

Second Semester

DESIGN AND ANALYSIS OF ALGORITHMS

(Common for M.C.A. (R) / WEP)

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. Define algorithm.

2. What is space complexity?

3. Define sequential search.

4. What is Greedy technique?

5. How to find the "degree of tree"?

6. What is the purpose of Dijkstra's algorithm?

7. Define breadth first search.

8. Define heap sort.

9. What is Hamilton's circuit? Give an example.

10. Define NP – complete problem.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

All questions carry equal marks.

11. (a) What are the important problem types? Explain them briefly.

Or

(b) Describe the desirable characteristic of an algorithm.

12. (a) Explain the Mathematical analysis of recursive algorithm.

Or

(b) Describe the Quick sort procedure with an example.

13. (a) Discuss the steps in developing a Dynamic programming algorithm. Illustrate with an example.

Or

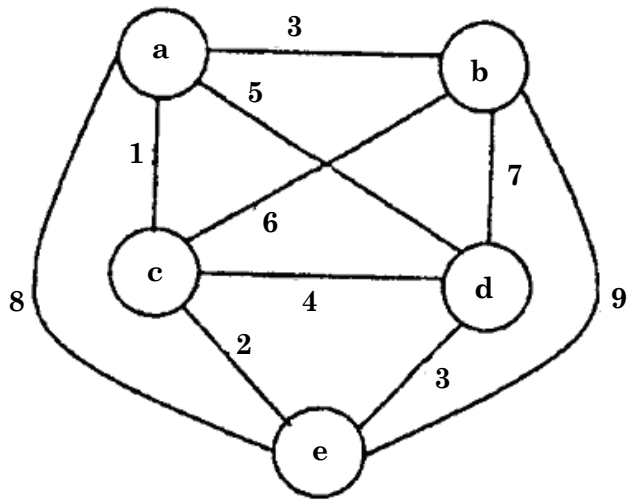
(b) Describe the Prim's algorithm.

14. (a) Explain the insertion sort with suitable example.

Or

(b) Describe the reduction of optimization problem.

15. (a) Apply the Branch and Bound algorithm to solve the travelling salesman problem :



Or

(b) Discuss about the NP complete problem.

Part C (3 × 10 = 30)

Answer any **three** questions.

All questions carry equal marks.

16. Explain Big Oh Notation, Omega Notation and Theta Notations. Give an examples.

17. Explain the Strassen's multiplication algorithm with suitable example.

18. Describe the Dijikstra's algorithm with suitable example.

19. Explain the concept of Heap sort with suitable example.

20. Explain the eight Queen's problem with an example.

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RW–10780

Sub. Code
541204/ 545204/ 551203

M.C.A. DEGREE EXAMINATION, APRIL 2016

Second Semester

COMMUNICATION SKILLS

(Common for M.C.A. (R)/M.C.A. (WEP)/ M.Sc. Computer Science)

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define extrapersonal communication.

2. What is called the process of information flowing from the lower levels of a hierarchy to the upper levels?

3. What is non-verbal communication?

4. What is the mode of greeting a person?

5. What is proximics?

6. How would you conclude a presentation?

7. What do you mean by team work?

8. What is the purpose of a job interview?

9. What does Cc and Bcc mean in an e-mail?

10. What is an agenda?

Part B $(5 \times 5 = 25)$

Answer each of the following questions, choosing either (a) or (b).

11. (a) Discuss the different channels of communication.

Or

(b) What are the barriers to communication?

12. (a) Explain the different modes of conversation.

Or

(b) Analyze the do's and don'ts of telephonic conversation.

13. (a) Highlight the significance of good delivery for a presentation.

Or

(b) Why do we audio-visual aids in presentation?

14. (a) Explain interview techniques.

Or

(b) Discuss the importance of group communication.

15. (a) How are résumés prepared?

Or

(b) How would you prepare the minutes of a meeting?

Part C $(3 \times 10 = 30)$

Answer any **three** of the following questions.

16. Examine the principles of effective communication.

17. Discuss the basic attributes of verbal and non verbal communication.

18. What are the characteristics of a good presentation?

19. What are the essential skills needed for Team Work?

20. Examine the format of a Report.

RW–10781

Sub. Code
541205/ 545205

M.C.A.–R/WE DEGREE EXAMINATION, APRIL 2016

Second Semester

ACCOUNTING AND FINANCIAL MANAGEMENT

(2012 onwards)

(Common for M.C.A. (R)/M.C.A. (WEP))

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is money measurement concept?

2. State any two limitations of trial balance.

3. What is Margin of safety?

4. What are the uses of cost sheet?

5. What is material variance?

6. What are the advantages of preparing sales budget?

7. What is profit maximization?

8. What is payback period?

9. What is stock dividend?

10. What is opportunity cost of capital?

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Prepare trading and profit and loss account from the information given below :

	Rs.
Opening stock	3,600
Purchases	18,260
Wages	3,620
Closing stock	4,420
Stock	32,000
Carriage on purchases	500
Carriage on sales	400
Rent (factory)	400
Rent (office)	500
Sales returns	700
Purchase returns	900
General expenses	900
Discount to customers	360
Interest from bank	200

Or

(b) Distinguish between trial balance and balance sheet.

12. (a) From the following information, prepare a summarised Balance sheet as at 31st March, 2014

(i) Working capital Rs. 120,0000

(ii) Reserve and surplus Rs. 80,000

(iii) Bank overdraft Rs. 20,000

(iv) Assets (Fixed)-proprietary Ratio 0.75

(v) Current ratio 2.5

(vi) Liquid ratio 1.5.

Or

(b) What is costing? Give its basic principles.

13. (a) Payoff Ltd. is producing articles mostly by manual labour and is considering to replace it by a new machine. There are two alternative models *M* and *N* of the new machine. Prepare a statement of profitability showing the pay-back period from the following information

	Machine *M*	Machine *N*
Estimated life of machine	4 years	5 years
Cost of machine	Rs. 9,000	Rs. 18,000
Estimated savings in scrap	Rs. 500	Rs. 800
Estimated savings in direct wages	Rs. 6,000	Rs. 8,000
Additional cost of maintenance	Rs. 800	Rs. 1,000
Additional cost of supervision	Rs. 1,200	Rs. 1800

Or

(b) Distinguish between standard cost and estimated cost.

14. (a) How do you justify profit-maximization objective of financial management. What are the defects of profit-maximization?

Or

(b) From the following information calculate Pay back period Initial investment Rs 20,000.

Year	Cash inflow (Rs.)
1	5,000
2	6,000
3	4,000
4	4,000
5	4,000

15. (a) What do you mean by optimum capital structure? Discuss briefly.

Or

(b) Velvet Corporation's outstanding preferred stock currently yields 11%. Velvet can issue new $100 par preferred with an 11% dividend, but will incur flotation costs of 5%. Find the cost of newly issued preferred stock.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. The Trial Balance of Mr. *X* as on 31st March 2015 was as follows :

	Dr. Rs.	Cr. Rs.
Purchases	1,62,505	
Sales		2,52,400
Reserve for doubtful debts		5,200
Sundry debtors	50,200	
Sundry creditors		30,526
Bills payable		3,950
Opening stock	26,725	
Wages	23,137	
Salaries	5,575	
Furniture	7,250	
Postage	4,226	
Power and fuel	1,350	

	Dr. Rs.	Cr. Rs.
Trade expenses	5,831	
Bad debts	525	
Loan at 10% pa. to Ram 1st Jan. 2015	3,000	
Cash on hand and at bank	10,000	
Outstanding wages		2,000
Trade expenses accrued but not paid		700
Drawings account	4,452	
Capital		10,000
	3,04,776	3,04,776

Prepare the Trading and P & L a/c for the year ended 31st March 2015 the balance sheet as on date taking into account the following information.

(a) Depreciation on furniture is to be charged at 10% p.a.

(b) Sundry debtors include an item of Rs. 500 due from a customer who has become insolvent.

(c) Reserve for doubtful debts is to be maintained at 5% on Sundry debtors.

17. What are the advantages and limitations of break even chart?

18. What are the various methods available for estimation of working capital requirements?

19. Balance sheet of a firm as on 31st March 2014 and 31st March 2014 were as follows :

Liabilities	31.3.2014 Rs.	31.3.2015	Assets	31.3.2014 Rs.	31.3.2015
Creditors	40,000	44,000	Cash	10,000	47,000
Loan from *X*	25,000	–	Debtors	30,000	50,000
Loan from *Y*	40,000	50,000	Stock	35,000	25,000
Capital	1,25,000	1,53,000	Machinery	80,000	55,000
			Land	40,000	10,000
			Building	35,000	60,000
	2,30,000	247,000		2,30,000	247,000

During the year a machine costing Rs. 10,000 (accumulated depreciation Rs. 3,000) was sold for Rs. 5,000. The balance of provision for depreciation against machinery as on 31st March 2014 was Rs. 25,000 and on 31st March 2015 Rs. 40,000. Net profit for the year 2014-15 amounted to Rs. 45,000. Prepare cash flow statement.

20. Explain the various types of dividend policy.

RW–10782

Sub. Code
541401/ 545401

M.C.A. (R)/(W.E.) DEGREE EXAMINATION, APRIL 2016

Fourth Semester

VISUAL PROGRAMMING WITH .NET

(Common for M.C.A. (R)/M.C.A. (WEP))

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. What is Visual studio?
2. What is a Web form?
3. What are the primitive data types available in VB.NET?
4. What is Method Snippets?
5. What is meant by Delegation Event model?
6. What is Manifest?
7. What is a Breakpoint?
8. Define Foreign Key. Give an example.
9. What are the features of VS – 2010?
10. What is a Layout?

Section B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Explain the methods to Navigate the Visual studio.

Or

(b) Explain the Database projects.

12. (a) Explain class Inheritance in VB.NET with example.

Or

(b) Explain any Two branching statements in VB.NET with examples.

13. (a) Explain Default Namespace with an example.

Or

(b) Explain how will you Navigating with Class view?

14. (a) Explain the Watch Windows.

Or

(b) Explain stored procedure with an example.

15. (a) Explain the Dock panel Layout with an example.

Or

(b) Explain how will you read and save data using Visual studio-2010?

Section C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain any Ten tools available in Visual studio.

17. Write a VB.NET program to demonstrate Parameter passing and Returning data.

18. Write a VB.NET program to add two complex numbers.

19. Explain the steps to create a Database.

20. Design a web page for your Department using suitable controls.

RW–10783

Sub. Code
541402/ 545402

M.C.A. DEGREE EXAMINATION, APRIL 2016

Fourth Semester

DISTRIBUTED COMPUTING

(CBCS 2012 onwards)

(Common for MCA (R)/MCA (WEP))

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. What are the advantages of distributed systems?

2. What is an internet?

3. What do you mean by marshalling?

4. What is the purpose of remote procedure call?

5. Define process.

6. Define cryptography.

7. What is logical time?

8. What is the need for synchronization?

9. Define transaction.

10. Write the conditions for distributed deadlock.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) What are the challenges in the distributed system?

Or

(b) Explain the internet protocols.

12. (a) Describe the client-server communication model.

Or

(b) Explain the communication between distributed objects.

13. (a) Write about digital signature.

Or

(b) Explain the distributed file system.

14. (a) Explain the DNS with an example.

Or

(b) Describe the distributed debugging.

15. (a) Explain optimistic concurrency control with an example.

Or

(b) Explain distributed multimedia system.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the various types of networks.

17. Explain the java RMI with an example.

18. Explain the cryptographic algorithms.

19. Describe the multicast communication related problems.

20. Explain flat and nested distributed transactions with example.

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RW–10784

Sub. Code
541403/ 545403

M.C.A. DEGREE EXAMINATION, APRIL 2016

Fourth Semester

WEB TECHNOLOGY

(CBCS – 2012 onwards)

(Common for MCA (R)/WEP)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. What is DOM?
2. What is a XML parser?
3. What are the advantages of Java Beans?
4. What is EJB?
5. What is the use of Tomcat Web Server?
6. What is JSDK?
7. What are the problems with Servlet?
8. What is meant by conditional processing?
9. What is JDBC?
10. Define Transaction.

Section B (5 × 5 = 25)

Answer **all** questions.

11. (a) Explain ordered list with an example.

Or

(b) Explain XML schemas with an example.

12. (a) Explain the design pattern for properties with example.

Or

(b) Explain various interfaces in Java Bean.

13. (a) Explain simple Servlet with an example.

Or

(b) Explain with an example how HTTP request can be handled?

14. (a) Explain the method in JSP with example.

Or

(b) Explain the implicit JSP objects.

15. (a) Explain the different types of JDBC drivers.

Or

(b) Explain any one Result Set with example.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Write a java script to calculate the Income Tax for the following.

Income	Tax Rate
Upto Rs. 3,00,000	No tax
Rs. 3,00,000 – 5,00,000	10%
Rs. 5,00,000 – 7,00,000	20%
Rs. 7,00,000	30%

17. Explain with an example how an EJB can refer another EJB?

18. Explain with an example how to handle HTTP GET request and HTTP POST request?

19. Explain with an example how a session attribute is created and how to read a session attribute by a JSP program?

20. Explain deploying Java Bean in a JSP page with an example.

RW–10785

Sub. Code
541552/ 545552

M.C.A.(R)/W.E. DEGREE EXAMINATION, APRIL 2016

Fourth Semester

Computer Application

SOFTWARE PROJECT MANAGEMENT

(2012 onwards)

(Common for M.C.A. (R)/M.C.A. (WEP))

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** the questions.

1. Write down the basic parameters of software Economics.

2. Mention any four general quality improvements with a modern process.

3. Define production stage.

4. Write any two primary objectives of inception phase.

5. Define Architecture baseline.

6. Define the term workflow.

7. Mention the issues of conventional work break down structures.

8. What are the three discrete states in the Project Environment?

9. What is Project Management?

10. Define Process.

Section B

$(5 \times 5 = 25)$

Answer **all** the questions.

11. (a) How to improve the software process in software management?

Or

(b) Write short notes on pragmatic software cost estimation.

12. (a) List out the principles of modern software management.

Or

(b) Explain the process of Engineering Artifacts.

13. (a) Write down the check points of the major mile stones process.

Or

(b) Write short notes on periodic status assessments.

14. (a) Mention the responsibilities of project organizations.

Or

(b) Write short notes on Automation Building Blocks.

15. (a) How to improve the Next generation software economics?

Or

(b) Write short notes on process maturing in tailoring the process.

Section C

(3 × 10 = 30)

Answer any **three** questions.

16. Describe in detail about the waterfall model.

17. Write a detailed note on the process of artifacts sets.

18. Explain in detail about the technical perspective of a software architecture.

19. Describe in detail about the iterative process planning guidelines.

20. Describe the management indicators in project control and process instrumentation.

RW–10786

Sub. Code
541561/545561

M.C.A (R)/(WE) DEGREE EXAMINATION, APRIL 2016

Fourth Semester

Computer Application

SOFT COMPUTING

(Common for MCA (R)/MCA (WEP))

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define learning process.

2. What do you mean by ANN?

3. What is back propogation network?

4. Define hope field network.

5. What is fuzzy set?

6. Define fuzzification.

7. Mention the uses of fuzzy measures.

8. Name the different modes of fuzzy approximate reasoning.

9. What is genetic algorithm?

10. What are the types of cross over?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Write down the applications of soft computing.

Or

(b) Write short notes on HEBB network.

12. (a) Write short notes on counter propagation network.

Or

(b) Explain ART network.

13. (a) Mention the properties of fuzzy sets.

Or

(b) Write short notes on fuzzy composition.

14. (a) Explain the formation of rules in fuzzy arithmetic.

Or

(b) Write short notes on aggregation of rules.

15. (a) Illustrate the application of genetic algorithm.

Or

(b) Write down the basic terminologies of genetic algorithm.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Describe in detail about neural network architecture.

17. Explain in detail about kohonen self organizing network.

18. Write a detailed note on defuzzification methods.

19. Discuss in detail about fuzzy inference and expert systems.

20. Describe in detail about the classification of genetic algorithm.

RW–10959

Sub. Code
541101/ 545101

M.C.A. DEGREE EXAMINATION, APRIL 2016

DIGITAL COMPUTER ORGANIZATION

(Common for MCA(R)/WE)

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Find the binary number for $(0.6875)_{10}$.

2. Write the binary equivalent for $(306.0)_{16}$.

3. Define flip flop.

4. What is meant by instruction code?

5. Write the definition of BCD.

6. Write any two register reference instructions with examples.

7. What are the storage components of CPU.

8. Define stack.

9. List the basic components of MMU.

10. Mention any four pheriparal devices.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Describe the fundamental concepts of Boolean algebra.

Or

(b) Write short notes on: Quick–Mccluskey method.

12. (a) Explain shift register in detail.

Or

(b) Write short notes on: 1's and 2's complements.

13. (a) Explain instruction cycle in detail.

Or

(b) Discuss about designing of basic computer.

14. (a) Briefly explain about various instruction formats.

Or

(b) Explain program control in detail.

15. (a) Write short notes on: DMA.

Or

(b) Discuss about virtual memory in detail.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. $F = (w, x, y, z) = \sum (1,4,5,6,7,13)$ .Simply by using K–map.

17. Discuss error detection codes.

18. Explain memory reference instructions.

19. Discuss addressing modes in detail

20. Explain asynchronous data tranfer.

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RW–10960

Sub. Code
541102/ 545102

M.C.A. (R)/WE DEGREE EXAMINATION, APRIL 2016

Computer Application

C AND DATA STRUCTURES

(2012 onwards)

(Common for MCA (R) / WE)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** the questions.

1. Define Constants.

2. What is meant by Keyword?

3. Define Array.

4. What are Function?

5. Define Data Structures.

6. What is meant by List?

7. Define Tree.

8. Mention the types of Binary Trees.

9. What is meant by Sorting?

10. What is the Best case Complexity of Linear Search?

Section B (5 × 5 = 25)

Answer **all** the questions, choosing either (a) or (b).

11. (a) What are Variables? Mention the Rules for defining Variables with examples.

Or

(b) Write short notes on Type conversions in C.

12. (a) What is meant by Recursion? Explain it with an Example.

Or

(b) Write a C program to illustrate the concept of "Array of Pointers".

13. (a) Write a short notes on: Singly Linked list.

Or

(b) Briefly explain Polish Notation with examples.

14. (a) Write a short notes on: Binary Tree Representations.

Or

(b) Describe about any TWO Hashing Techniques.

15. (a) Briefly explain about Binary Search.

Or

(b) Elaborate Quick Sort with example.

Section C

(3 × 10 = 30)

Answer any **three** questions.

16. Explain Formatted & Unformatted I/O Functions with examples.

17. Describe any FIVE String Handling Functions in C with Examples.

18. Discuss about Implementation of Stack.

19. Explain Insertion & Deletion operations in Binary Search Tree with examples.

20. Write short notes on:

    (a) Bubble Sort

    (b) Selection Sort.

RW–10961

Sub. Code
541103/ 545103

M.C.A. DEGREE EXAMINATION, APRIL 2016

Comp. Applications

RELATIONAL DATABASE SYSTEMS

(Common for M.C.A.(R)/WE)

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** the questions.

1. Define: DDL.
2. What is meant by Storage Manager?
3. Define: Integrity Constraint.
4. Write a note on Division operation in Relational Algebra.
5. What are Nested SQL queries?
6. Define Second Normal Form.
7. List the Transaction states.
8. Define: Serializable schedules.
9. What is meant by Cluster index?
10. State any two differences between Primary index and Secondary index.

Part B $(5 \times 5 = 25)$

Answer **all** the questions, choosing either (a) or (b).

11. (a) Describe the View's of data in Database.

Or

(b) Write short notes on: Transaction Management.

12. (a) Describe about logical database design.

Or

(b) Write short notes on: Projection and Join operations in Relational Algebra.

13. (a) List any FIVE aggregative operations in SQL and explain them with examples.

Or

(b) Discuss about Lossless Join Decomposition.

14. (a) Describe about Testing for Serializability.

Or

(b) Write short notes on: Buffer Management.

15. (a) Write short notes on: Tree based indexing.

Or

(b) Write a short note on: B^+ trees.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain the Database System Structure.

17. Explain Tuple and Domain Relational Calculus with examples.

18. Discuss Multi valued dependencies with examples.

19. Explain about Timestamp - based protocols.

20. Discuss in detail, Comparison of file organizations.

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RW–10962

Sub. Code
541104/ 545104

M.C.A. (R/WE) DEGREE EXAMINATION, APRIL 2016

Computer Applications

DISCRETE MATHEMATICS

(CBCS – 2012 onwards)

(Common for M.C.A. (R)/WE)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. What is Principle Disjunctive Normal Form?

2. Construct the Truth table for $P \wedge \rceil P$

3. Prove the identity by Set theory
   (a) $A \cap E = A$
   (b) $A \cup E = E$.

4. Give an example of a relation which is neither reflexive nor irreflexive.

5. What are Semi group and Submonoids?

6. What is called a Ring?

7. Write a note on Trees. Explain degree of the node.

8. What is null graph? Explain with a diagram.

9. What is Exhaustive event? Explain with an example.

10. What is the Probability of having a jack and a queen when two cards are drawn from a pack of 52?

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Show that
$P \to (Q \to R) \Leftrightarrow P \to (\neg Q \vee R) \Leftrightarrow (P \wedge Q) \to R$.

Or

(b) Construct the Truth table for the formula : $\neg(P \wedge Q) \leftrightarrow \neg P \vee \neg Q$

12. (a) Let $X = \{1, 2, 3, ..., 7\}$ and $R = \{(x, y) \mid x - y$ is divisible by 3}. Show that R is an Equivalence relation. Draw the graph of R.

Or

(b) Let $R = \{(1, 2), (3, 4), (2, 2)\}$
and $S = \{(4, 2), (2, 5), (3, 1), (1, 3)\}$.
Find $R \circ S$, $S \circ R$, $R \circ (S \circ R)$, $(R \circ S) \circ R$. $R \circ R$, $S \circ S$, and $R \circ R \circ R$.

13. (a) Show that the Intersection of two normal Subgroup is a Normal Subgroup.

Or

(b) Show that $(\{1, 4, 13, 16\}, X17)$ is a sub group of (Z_{17}^{*}, X_{17}).

14. (a) Prove that in an acyclic simple digraph a node base consist of only those nodes whose indegree is zero.

Or

(b) Give a directed tree representation of the following formula.
$(P \vee (\neg P \wedge Q)) \wedge ((\neg P \vee Q) \wedge \neg R)$

15. (a) If A, B, C are mutually independent events then $A \cup B$ and C are Independent. Prove.

Or

(b) A bag contains 6 green, 5 yellow and 7 white balls 4 balls are drawn at random from a bag. Find the probability that among the balls drawn, there is at least one ball of each color.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Show that $((P \vee Q) \wedge \neg(\neg P \wedge (\neg Q \vee \neg R))) \vee (\neg P \wedge \neg Q) \vee (\neg P \wedge \neg R)$ is a Tautology.

17. Let $X = \{2, 3, 6, 12, 24, 36\}$ and the relation $\leq$ be such that $X \leq Y$ if X divides Y. Draw the Hasse diagram of $(X, \leq)$.

18. Let $(S, *)$ be a semi group and Z€S be a left zero. Show that any X € S, $X * Z$ is also a left zero.

19. Give a directed tree representation of the following formula $P \vee (\neg P \wedge Q) \wedge ((\neg P \vee Q) \wedge \neg R)$. From this representation obtain the corresponding prefix formula.

20. If X is the number of points rolled with a balanced die, find the expected value of X.

RW–10963

Sub. Code
541303/ 545303

M.C.A (R)/WE DEGREE EXAMINATION, APRIL 2016

COMPUTER NETWORKS

(Common for M.C.A.(R)/WE)

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Computer Network.
2. What is Signal to Noise Ratio?
3. What are the functions of data link layer?
4. What is flow control? Mention it types.
5. Define flooding.
6. Define Broadcast.
7. What is meant by Remote Procedure Call?
8. Define browser.
9. What is symmetric key cryptography?
10. Define cipher text with an example.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Describe the need of Fiber optical cable.

Or

(b) Explain coaxial cable.

12. (a) Explain Go-back n ARQ.

Or

(b) Explain CSMA/CA.

13. (a) What is Packet Switching? Explain.

Or

(b) Explain Message Switching.

14. (a) Write about TCP.

Or

(b) Explain about HTTP.

15. (a) Explain transposition ciphers technique in cryptography.

Or

(b) Describe the AES Algorithm.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the categories of Networks. Describe.

17. Explain Error detection and correction techniques in data link layer.

18. Explain the shortest path routing algorithm with an example.

19. Explain in detail about electronic mail architecture and its transmission technologies.

20. Explain DES cryptographic technique with an example.

RW–10964

Sub. Code
541560/ 545560

M.C.A. (R)/WE DEGREE EXAMINATION, APRIL 2016

Mathematics

Elective — RESOURCE MANAGEMENT TECHNIQUES

(2012 onwards)

(Common for M.C.A. (R)/WE)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. How will you overcome degeneracy in transportation problem?

2. What are the methods to use for solving an integer programming problem?

3. Explain the term in PERT: Three time estimates?

4. What are the characteristic of LPP?

5. What is Branch and Bound Technique?

6. What is the purpose of Fractional cut constraints?

7. What are the advantages of CPM and PERT techniques?

8. Define total float and free float.

9. List any two applications of Queuing theory.

10. Explain the terms 'jockeying' and reneging' in queuing models.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Use simplex method to solve the LPP

Maximize $Z = 3x_1 + 5x_2$

Subject to the constraints

$$3x_1 + 2x_2 \le 18$$
$$0 \le x_1 \le 4$$
$$0 \le x_1 \le 6.$$

Or

(b) Use a Two-Phase simplex method to

Maximize $Z = 3x_1 + 2x_2$

Subject to the constraints

$$2x_1 + x_2 \le 2$$
$$3x_1 + 4x_2 \ge 12$$
$$x_1, x_2 \ge 0.$$

12. (a) Solve the following assignment problem

	I	II	III	IV	V
A	10	5	9	18	11
B	13	19	6	12	14
C	3	2	4	4	5
D	18	9	12	17	15
E	11	6	14	19	10

Or

(b) Write in detail about the Transportation problem with an example.

13. (a) What are the methods to use for solving an integer programming problem?

Or

(b) Solve the following all-Integer programming problem using the branch and bound method.

Minimize $Z = 3x_1 + 2.5x_2$

Subject to the constraints

$$x_1 + 2x_2 = 20$$
$$3x_1 + 2x_2 = 50.$$

14. (a) Discuss the different phases of Project scheduling.

Or

(b) Explain the function of Probability and Cost consideration in Project scheduling.

15. (a) Explain about ABC analysis in detail.

Or

(b) Explain in detail about the Queuing theory.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Solve the given linear programming problem using simplex algorithm method

Maximize $Z = 4x_1 + 5x_2$

Subject to

$$2x_1 + 3x_2 \leq 8$$
$$4x_1 + 4x_2 \leq 10$$
$$x_1 \text{ unrestricted}, x_2 \geq 0.$$

17. Solve the following problem using Gomory's cutting plane method.

Maximize $f = 6x_1 + 7x_2$

Subject to

$$7x_1 + 6x_2 \le 42$$
$$5x_1 + 9x_2 \le 45$$
$$x_1 - x_2 \le 4$$
$$x_i \ge 0 \text{ and integer, } i = 1,2.$$

18. Solve the following integer programming problem using Branch and Bound technique

Maximize $z = 10x + 20y$

Subject to :

$$6x + 8y \le 48$$
$$x + 3y \le 23$$
$$x, y \ge 0 \text{ and integers.}$$

19. Discuss the types of float in CPM and bring out the difference between PERT and CPM.

20. What is Inventory Management and how EOQ is maintained with uniform demand and shortages?

RW–10965

Sub. Code
541551/545551

M.C.A. (R/WE) DEGREE EXAMINATION, APRIL 2016

Third Semester

OBJECT ORIENTED ANALYSIS AND DESIGN

(Common for M.C.A. (R/WE)

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Class.

2. Define Object.

3. What is Static Model?

4. Define Rumbaugh Object Modelling Technique.

5. Define Analysis.

6. What is process in Object Oriented Analysis?

7. What is a Conceptual Model?

8. Define Polymorphism.

9. What is evaluation testing?

10. Write any four OOD activities.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Describe the relationships among objects.

Or

(b) Explain Multiple Inheritance with an example.

12. (a) Write about pattern.

Or

(b) Describe in brief Unified Modeling Language.

13. (a) Explain about Documentation.

Or

(b) Describe in brief about Transactions.

14. (a) Elucidate about Designing axioms.

Or

(b) Write the characteristics of OOD.

15. (a) Write the merits and demerits of black box testing.

Or

(b) Define: System Usability and Measuring User satisfaction.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the following:
    (a) Prototyping
    (b) Rapid Application Development.

17. Explain the Booch Methodology.

18. Write in detail about Classification.

19. Elucidate in detail about Designing Classes.

20. Write in detail about Software Quality Assurance.

RW–11036

Sub. Code
541562/ 545562

M.C.A. R/WE DEGREE EXAMINATION, APRIL 2016

Computer Applications

GRID COMPUTING

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. What are the key issues of grid computing?

2. What is the difference between P_2P and grid computing?

3. Give any two examples for news scientific and engineering applications enabled by grid environments.

4. What are the major phases in computing and communications?

5. What is the use of data grid?

6. What is SOAP?

7. Write the objectives of OGSA.

8. List out the properties of grid services.

9. Give any two security considerations.

10. What are the platform services provided by OGSA?

Part B $(5 \times 5 = 25)$

Answer **all** the questions.

11. (a) Describe the different types of grid.

Or

(b) Write a note on virtualization.

12. (a) Describe the evolution of grids over time.

Or

(b) What are the types of tools available and explain them briefly?

13. (a) Describe the role of scheduler functional block.

Or

(b) Write about web services resource framework.

14. (a) Describe the benefits of OGSI standard.

Or

(b) Write the features of SOAP.

15. (a) Describe the security requirements.

Or

(b) Describe the interfaces that can be used to construct a variety of VO service structures.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the potential applications and financial benefits of grid computing.

17. Describe the challenges of grid computing.

18. Describe the service view of grid computing.

19. Draw the structure of OGSA architected services and explain.

20. What are the principal elements of OGSA and explain?

RW–10777

Sub. Code
541201/ 545201

M.C.A. DEGREE EXAMINATION, APRIL 2016

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(Common for M.C.A. (R)/WEP)

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What do you mean by Dynamic Binding?

2. What is the use of Manipulators in C++?

3. What is meant by Default Constructor?

4. What are the advantages of inline function?

5. What is an Indirect Base class? Give an example.

6. What is the need for virtual function?

7. What is a Template?

8. What is the difference between Binary file and an ASCII file?

9. Define Exception.

10. What is meant by fault tolerant design?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Explain the basic Concepts of OOP.

Or

(b) Explain the unformatted I/O operations with example.

12. (a) Write a C++ program to demonstrate call by reference.

Or

(b) Explain array of pointers with an example.

13. (a) Explain multiple inheritance with a program.

Or

(b) Explain pure virtual function with a program.

14. (a) Explain class template with an example.

Or

(b) Explain various file handling functions in C++ with example.

15. (a) Explain the basic exception handling mechanism in C++ with an example.

Or

(b) Explain the memory allocation failure exceptions with an example.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Write a C++ program using *while loop* to calculate the sum of the numbers that are divisible *5* and *7* and not divisible by *9* between *n1* and *n2*.

17. Write a C++ program to calculate the biggest of two numbers in two different classes using friend function.

18. Write a C++ program to convert given rupee into dollar using type conversion between objects and basic types.

19. Write a C++ program to swap two integers, two float numbers and two characters using function template.

20. Explain with a program how will you handle exception in operator overloaded functions.